Antonio Pelusi

PhD student in Computer Science and Mathematics of Computation

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Summary

Research focused on Differential Privacy and parameter-efficient fine-tuning of LLMs, with papers on tabular data synthesis, memorization risks, and explainable governance of synthetic health data.

Engineering experience in Python and C, covering ML pipelines, evaluation frameworks, and high-throughput data platforms.

Education

PhD in Computer Science and Mathematics of Computation | Nov 2024 - Oct 2027

University of Insubria (Varese, Italy), Random Power (Remote), IBM Research (Dublin, Ireland)

• Researched privacy-preserving fine-tuning of locally deployable Large Language Models (LLMs) for synthetic tabular data generation.

• Investigated quantum entropy noise sources in Differential Privacy mechanisms, characterizing theoretical privacy budgets and empirical protection.

Master's Degree in Computer Science | Sep 2022 - Apr 2024

University of Modena and Reggio Emilia (Modena, Italy) | Grade: 110/110

• Curriculum in Machine Learning, Cryptography, and Distributed Systems. Master's thesis on Differential Privacy for tabular data sharing.

Postgraduate Course CBI.Attract | Feb 2023 - Jun 2023

University of Bologna (Bologna, Italy), CERN (Geneva, Switzerland), AGH University (Kraków, Poland) | Grade: 30/30 with Honors

• Prototyped emerging technology solutions via Design Thinking within the EU ATTRACT research framework.

Bachelor's Degree in Computer Science | Sep 2018 - Apr 2022

University of Modena and Reggio Emilia (Modena, Italy), Liferay Inc. (Remote)

• Foundations in algorithms, complexity, software engineering, and mathematics.

Experience

Visiting Researcher | Sep 2026 - Mar 2027

IBM Research (Dublin, Ireland)

• Conducted joint research on differentially private fine-tuning of foundation models for structured data synthesis in enterprise environments.

Software Developer (Part-time) | May 2024 - Nov 2025

Random Power (Remote)

• Built a C Entropy-as-a-Service platform compliant with the Palo Alto Networks QRNG API, streaming quantum random bits at 1 Gbit/s.

• Developed a full-stack Django management interface for the 64x Multi-Generator Board, implementing hardware-software communication protocols and session authentication.

Publications

Towards Explainable Governance of Synthetic Health Data in Collaborative Healthcare Ecosystems | Oct 2026

A. Pelusi (University of Insubria, Varese, Italy), A. Trombetta (University of Insubria, Varese, Italy), D. Spoladore (CNR, Milan, Italy), S. Braghin (IBM Research, Dublin, Ireland) | PRO-VE 2026, Padua, Italy

• Proposed a role-aware Explainable AI framework for the governance of synthetic health data, mapping privacy and fidelity metrics to stakeholder-specific requirements under GDPR and the EU AI Act.

Privacy-Fidelity Trade-offs in Differentially Private Synthetic Tabular Data Generation | Jul 2026

A. Pelusi (University of Insubria, Varese, Italy), S. Braghin (IBM Research, Dublin, Ireland), A. Trombetta (University of Insubria, Varese, Italy) | Under review

• Evaluated Low-Rank Adaptation (LoRA) fine-tuning of LLMs with DP-SGD via Opacus for tabular synthesis, identifying privacy-induced degradation on high-cardinality categorical distributions.

Categorical Prior Lock-in: Why In-Context Learning Fails for Structured Data | Apr 2026

A. Pelusi (University of Insubria, Varese, Italy), S. Braghin (IBM Research, Dublin, Ireland), A. Trombetta (University of Insubria, Varese, Italy) | Under review

• Compared in-context learning against parameter-efficient fine-tuning for tabular data synthesis, characterizing the fidelity-memorization trade-off in foundation models.

Projects

synth-pipeline | Oct 2026

Privacy-Preserving Tabular Data Synthesis Pipeline

• Built a configurable pipeline for synthetic tabular data generation with LoRA-fine-tuned LLMs, with optional Differential Privacy (DP-SGD via Opacus) and schema-validated outputs.

• Implemented an automated benchmark suite assessing statistical fidelity (TVD, correlation error), machine learning utility (TSTR), and privacy protection (distance to closest record, membership inference).

chesSLM | Jun 2026

Decoder-only Language Model Trained on Chess Games

• Engineered a ~100M-parameter Transformer from scratch in PyTorch featuring Pre-LN blocks, Rotary Positional Embeddings (RoPE), and FlashAttention kernels.

• Trained via a two-stage curriculum (mixed precision, gradient accumulation, cosine decay) and packaged as a standalone UCI chess engine rated ~1650 Elo.

epsiloc | Oct 2025

Local Differential Privacy Library for Tabular Data

• Implemented vectorized Laplace and randomized response perturbation algorithms in Python for local differential privacy enforcement on structured datasets.

Awards

Winner of Green & Digital Transformation Hackathon | Sep 2023

University of Modena and Reggio Emilia (Mantua, Italy)

• Developed an XGBoost predictive model for municipal water distribution networks, winning first place among 15 competing teams.

Skills

Programming languages: Python, C, C++, Java

Research areas: Differential Privacy, Synthetic Data Generation, LLM Fine-Tuning, Membership Inference, Explainable AI

ML/AI libraries: PyTorch, Hugging Face Transformers, PEFT, Opacus, scikit-learn, NumPy, pandas

Web: HTML, CSS, JavaScript, Django

Tools: Linux, Bash, Git, Docker, LaTeX

Spoken languages: Italian (native), English (B2)

Other Experience

Computer Science High School Teacher (Part-time) | Sep 2024 - Jul 2025

A. Paradisi High School (Vignola, Italy)

• Taught programming principles, data structures, and algorithms to 75+ secondary school students.

Computer Science University Tutor (Part-time) | Sep 2024 - Dec 2024

University of Modena and Reggio Emilia (Modena, Italy)

• Mentored 50+ undergraduate students in laboratory sessions on programming and data structures.

Junior Software Developer (Internship) | Mar 2021 - Jul 2021

Liferay Inc. (Remote)

• Implemented Stripe payment integration modules for the open-source enterprise portal in Java.